CMPE2150 PCB Project X

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> **!** Important
>
> You should already have completed an exercise introducing and exploring KiCAD 9.0 and worked on the CMPE2150 PCB Project. This project is a similar but smaller version of the that one, suitable for a skills check or timed exam. It assumes your familiarity with those exercises and notes and will therefore be comparatively terse. See the original project if you need more clarity on specific tasks. The steps in this exercise are broadly the same and presented in the same order as the original project.

Part 1 - Schematic Capture

Your first

The Circuit Design—Programmable Pulse Generator

The circuit we have selected for you to realize is shown below. You should also have access to an NI Multisim file for the circuit, which will allow you to see it in detail and even simulate it to investigate its behaviour.

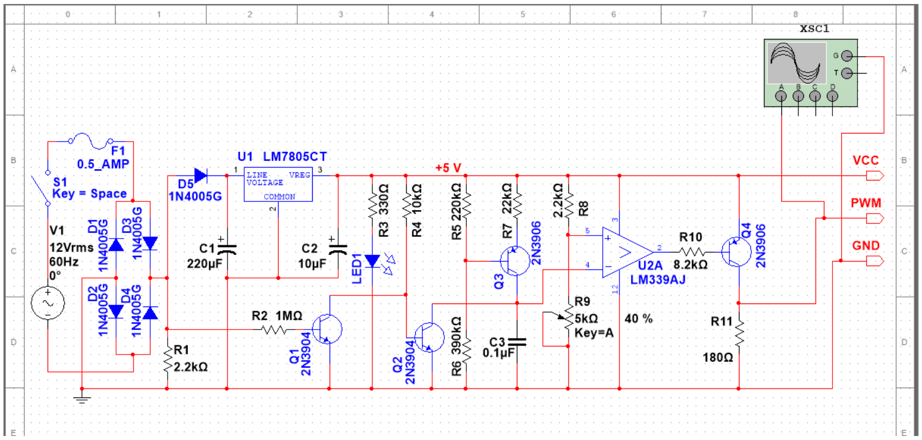

Figure 1: Programmable Pulse Generator

This circuit is from the classic *The Art of Electronics* [1, p. 242]. It is a variable width single pulse generator. The potentiometer sets the duration of the pulse and the switch triggers it.

The four-pin header represents the board's inputs and outputs:

- V_S is the voltage source to power the op amps (the top rail). It will ordinarily be $+15V_{DC}$.
- V_{cc} is the voltage source for the 555 timer and also serves as the "on" digital voltage reference. It will be $+5V_{DC}$.
- *Out* is the output of the timer. This is where the pulse generated by the circuit appears.
- *Gnd* is the ground reference for the other three pins.

A - KiCAD Project Setup

Follow the procedure below—nearly identical to the detailed one presented in the tutorial—to create your project to realize the schematic above.

1. Open KiCAD 9.0 and select *File* → *New Project*. Name your project "PulseGenerator-<*UserID*>", replacing <*UserID*> with your NAIT username. You do not need to create a new folder if you already did so for the Github project. Then click *Save*
2. If you haven't already done so, setup the Symbol Library to use the recommended option of copying the default global symbol library table.
3. For settings not mentioned above, select the defaults or use your own judgement.

[1] P. Horowitz and W. Hill, *The art of electronics*, 3rd ed. Cambridge University Press, 2015.

B - Schematic Sheet Setup

Now, open the *Schematic Editor* and perform the following procedure:

1. Select *File → Page Settings* to open the sheet setup.
2. Choose a size of *8.5 x 11in* with *Landscape* orientation. Set the revision to 0. Use today's date. You'll only need one sheet. Add a title of "CMPE2150 PWM Generator - <*UserID*>[1]". Use "NAIT CNT" as your Company. Add your full name to the first comment. Save your work and confirm that the *Title Block* looks correct.

[1] *Mutatis mutandis*

C - Place Schematic Symbols

It's finally time to start capturing our schematic. We often do this in several phases: first we will place the component symbols using the NEMA (US) symbols that you are used to. Then we will wire the schematic with connections between the components. This will be followed by choosing the values, features and appropriate footprints for our PCB realization. Finally, we will make some adjustments to the schematic to prepare it for use in creating our PCB design. Of course, as you become familiar with the tools and processes, you won't be quite so fastidious. It is common to select values and footprints, and even wire components together as we place them. While we won't stop you from doing so, beginners may find it helpful to go through this more structured phased approach.

Add the components from our schematic using the tools and techniques you learned from the tutorial. The notes below may be helpful.

1. Get used to using the hotkeys from the KiCAD cheat sheet rather than selecting buttons and menus (eg. using 'A' for adding a symbol). It's much faster and more efficient once you get used to them.
2. Use the grid to align components neatly, anticipating the upcoming wired connections.
3. At this point, focus on neat placement and arrangement of the symbols. Do not worry overmuch about values and footprints until later.
4. Please use NEMA (sometimes called 'US' or 'American') symbols. The only one likely to matter is the resistors. Use the sawtooth symbol, not the rectangle (look for `_US` in the symbol name).
5. You can use a `CONN_01x04`[2] to get the three output pins on the left of the schematic.

[2] Four-pin header row

The components in the schematic are listed below, along with a description of the component part number and footprint you should use.

Part 4 - Wire the Schematic

Once you have all the symbols (or suitable stand-ins) in place. Use the techniques and tools that you used in the tutorial. This is pretty straightforward, but neatness counts. Ensure that symbols are properly aligned and rotated, and that the wire routes are short, simple and neat.

Notes:

1. Add a `VCC` and `GND` net connection using power symbols (P), as described in the tutorial. Note that the AC source still *does not* connect to those nets. That's one thing we'll deal with in the final stage.
2. Let's also add some power symbols (P) for the AC power side. Use `VAC` on the top of the AC source the way you used `VCC` above, and `GNDPWR` for the bottom.
3. Add power flags (`PWR_FLG`) too all four of those for now.
4. Visually confirm that you have all the connections in the original schematic.
5. Again, remember to check in and push when you're done this stage. Remember to update the revision number.

Submission

When you have completed the above and checked and reviewed your work, increment the version number (yet again), check in and push the schematic design. Ask your instructor to review your schematic with you, and make any changes they require. When you are done (after, of course, incrementing the version number, checking in and pushing) submit the schematic.